

EXPERIMENT 27

Permission Auditing

AIM

To examine file permissions and identify files with excessive read, write, or execute permissions without modifying them.

ALGORITHM

1. Ask for a directory.
2. Read the permissions of each file.
3. Convert permissions into a readable form.
4. Flag files writable by everyone or executable when unexpected.
5. Assign a simple risk level.
6. Display the permission causing the warning.

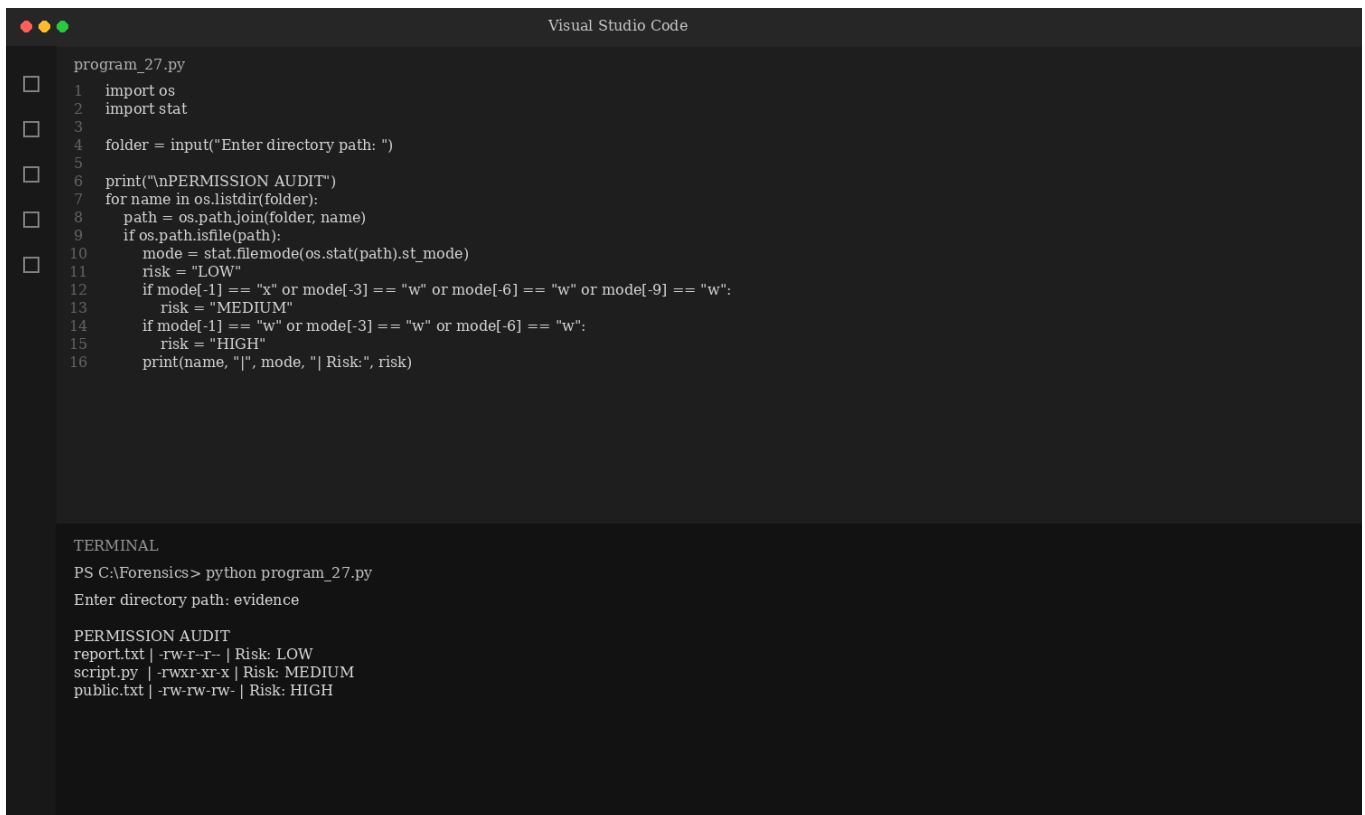
CODE

```
import os
import stat

folder = input("Enter directory path: ")

print("\nPERMISSION AUDIT")
for name in os.listdir(folder):
    path = os.path.join(folder, name)
    if os.path.isfile(path):
        mode = stat.filemode(os.stat(path).st_mode)
        risk = "LOW"
        if mode[-1] == "x" or mode[-3] == "w" or mode[-6] == "w" or mode[-9] == "w":
            risk = "MEDIUM"
        if mode[-1] == "w" or mode[-3] == "w" or mode[-6] == "w":
            risk = "HIGH"
        print(name, "|", mode, "| Risk:", risk)
```

OUTPUT



The screenshot shows the Visual Studio Code editor with a Python script named `program_27.py` and its terminal output. The script is a permission auditing tool that takes a directory path as input and lists files with their permissions and a risk level. The terminal shows the execution of the script on the `evidence` directory, resulting in a list of files and their risk levels.

```
program_27.py
1  import os
2  import stat
3
4  folder = input("Enter directory path: ")
5
6  print("\nPERMISSION AUDIT")
7  for name in os.listdir(folder):
8      path = os.path.join(folder, name)
9      if os.path.isfile(path):
10         mode = stat.filemode(os.stat(path).st_mode)
11         risk = "LOW"
12         if mode[-1] == "x" or mode[-3] == "w" or mode[-6] == "w" or mode[-9] == "w":
13             risk = "MEDIUM"
14         if mode[-1] == "w" or mode[-3] == "w" or mode[-6] == "w":
15             risk = "HIGH"
16         print(name, "|", mode, "| Risk:", risk)
```

TERMINAL

```
PS C:\Forensics> python program_27.py
Enter directory path: evidence

PERMISSION AUDIT
report.txt | -rw-r--r-- | Risk: LOW
script.py | -rwxr-xr-x | Risk: MEDIUM
public.txt | -rw-rw-rw- | Risk: HIGH
```

RESULT

The program audits file permissions and assigns a simple risk level to each file.